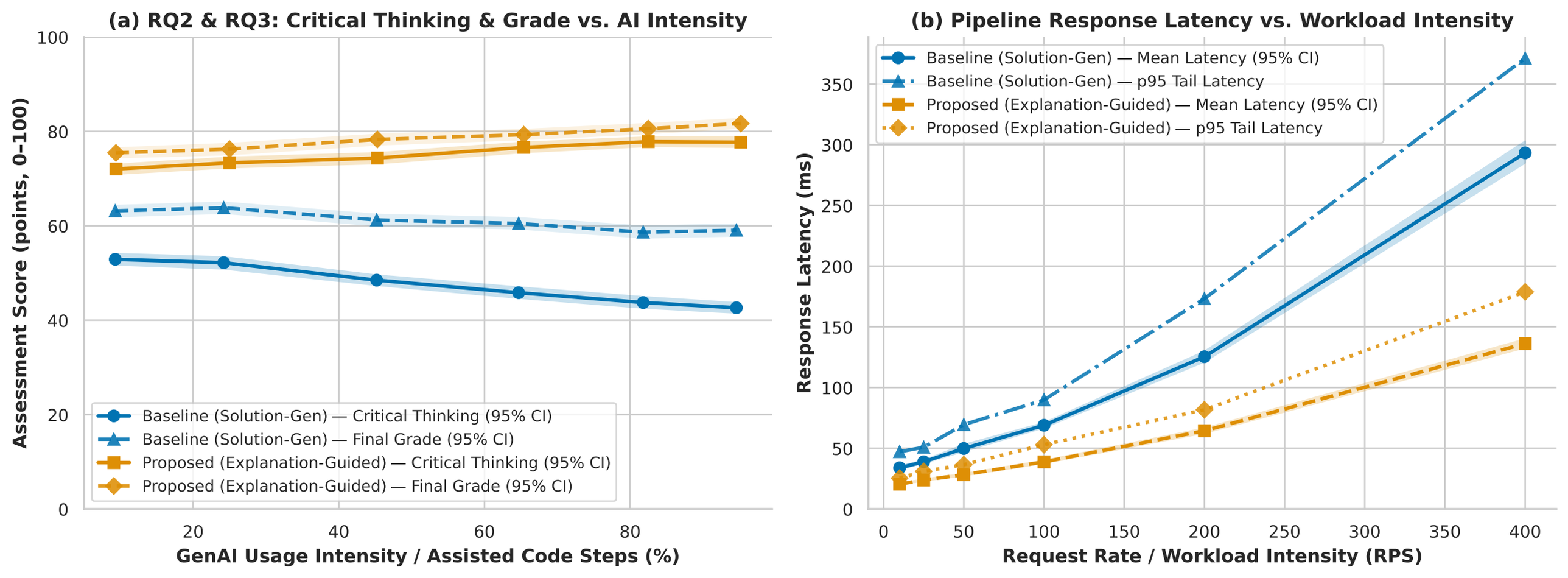
(a) RQ2 & RQ3: Critical Thinking & Grade vs. AI Intensity
Assessment Score (points, 0–100)
GenAI Usage Intensity / Assisted Code Steps (%)
Baseline (Solution-Gen) — Critical Thinking (95% CI)
Baseline (Solution-Gen) — Final Grade (95% CI)
Proposed (Explanation-Guided) — Critical Thinking (95% CI)
Proposed (Explanation-Guided) — Final Grade (95% CI)
(b) Pipeline Response Latency vs. Workload Intensity
Response Latency (ms)
Request Rate / Workload Intensity (RPS)
Baseline (Solution-Gen) — Mean Latency (95% CI)
Baseline (Solution-Gen) — p95 Tail Latency
Proposed (Explanation-Guided) — Mean Latency (95% CI)
Proposed (Explanation-Guided) — p95 Tail Latency